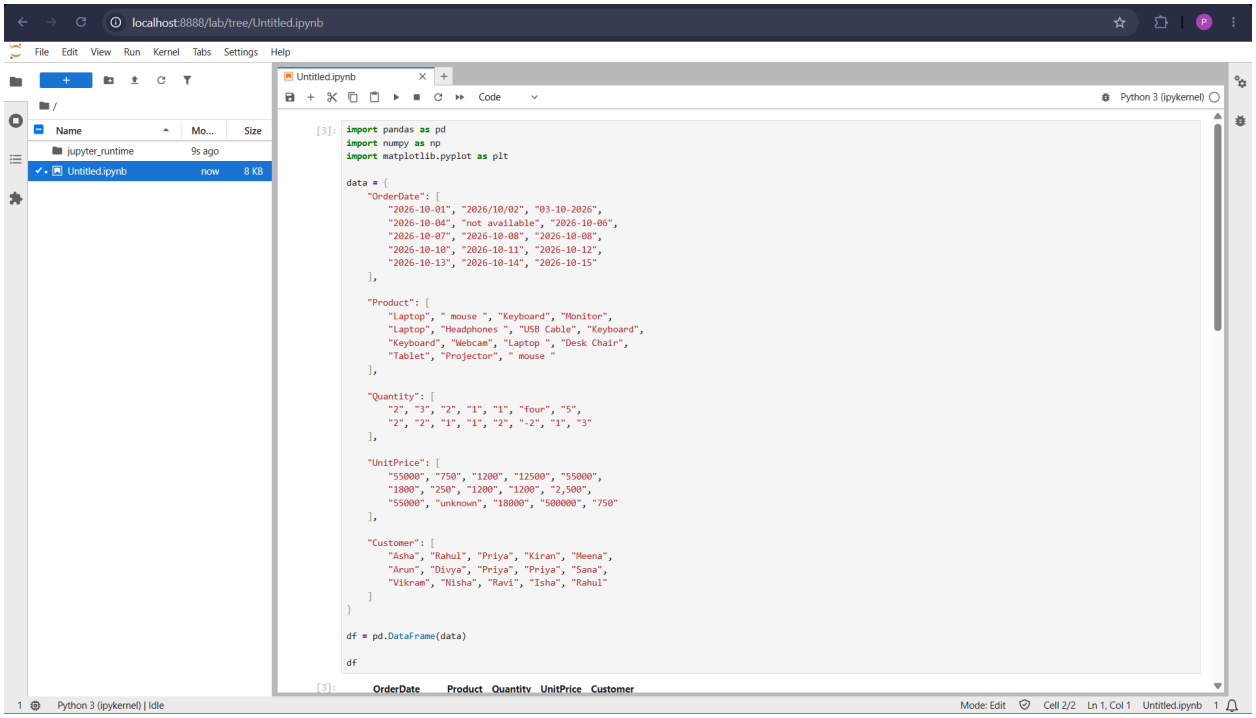
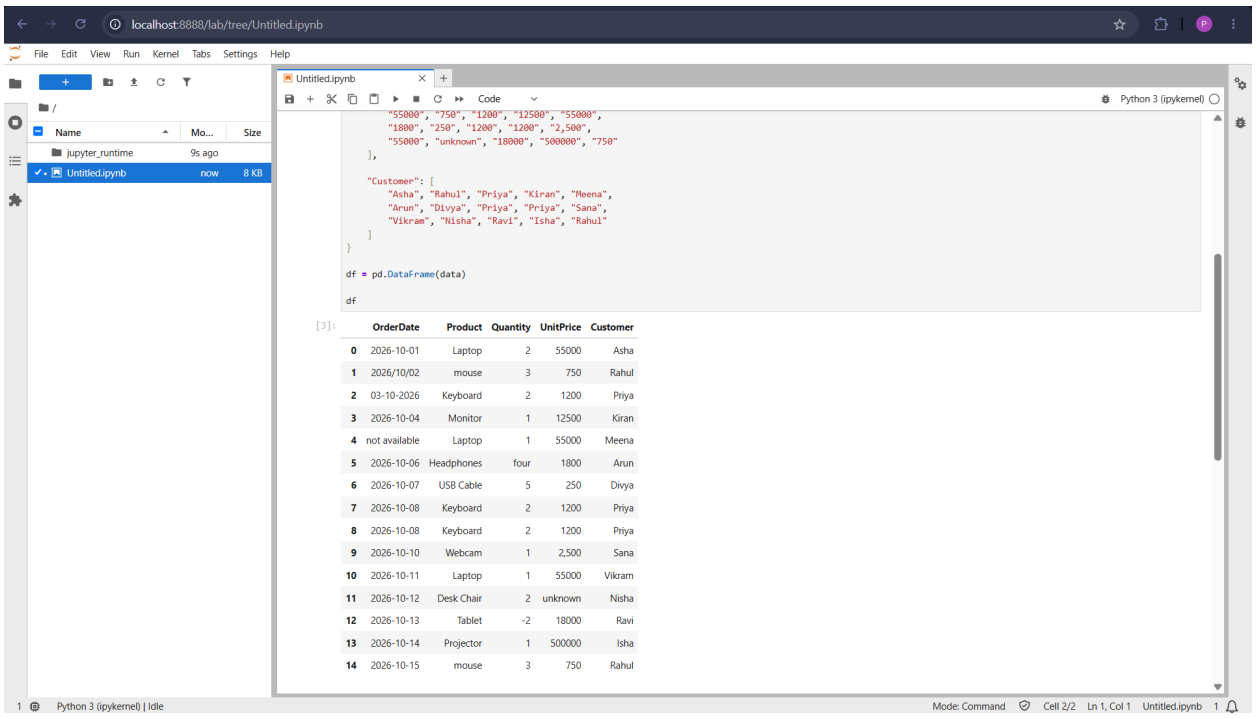
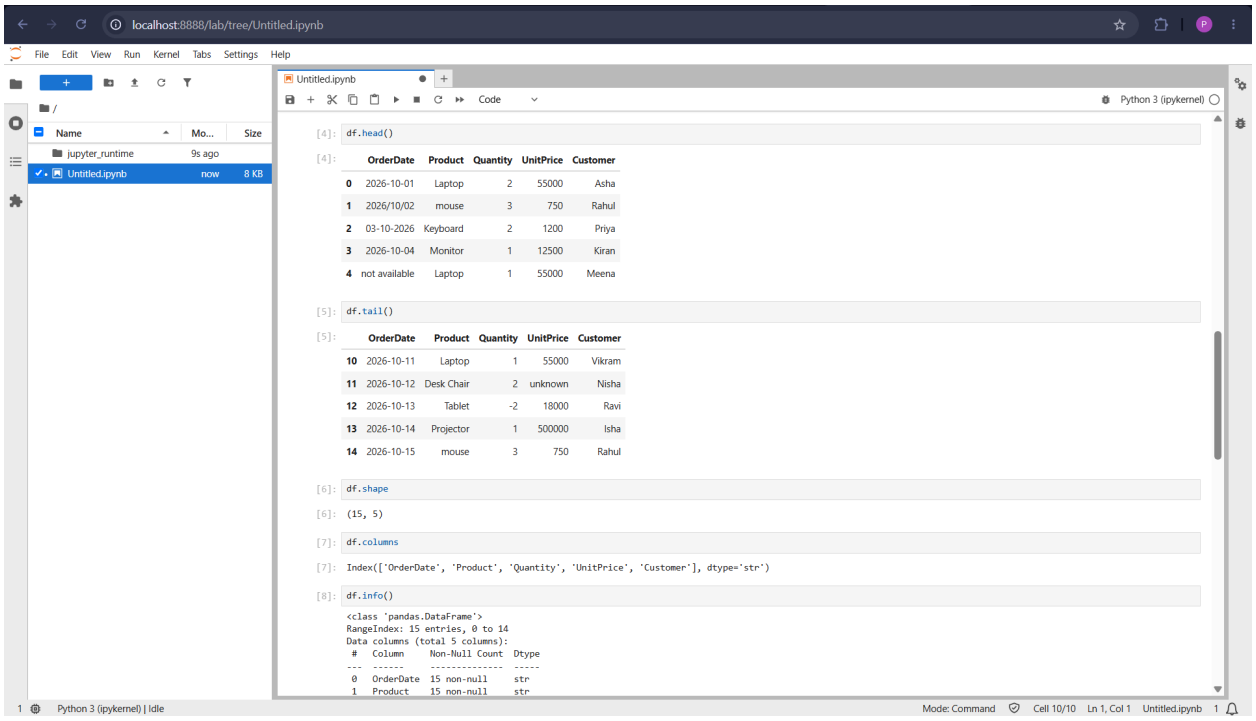

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Untitled.ipynb
Code
Python 3 (ipykernel)
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[3]:
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
data = {
"OrderDate": [
"2026-10-01", "2026/10/02", "03-10-2026",
"2026-10-04", "not available", "2026-10-06",
"2026-10-07", "2026-10-08", "2026-10-08",
"2026-10-10", "2026-10-11", "2026-10-12",
"2026-10-13", "2026-10-14", "2026-10-15"
],
"Product": [
"Laptop", " mouse ", "Keyboard", "Monitor",
"Laptop", "Headphones ", "USB Cable", "Keyboard",
"Keyboard", "Webcam", "Laptop ", "Desk Chair",
"Tablet", "Projector", " mouse "
],
"Quantity": [
"2", "3", "2", "1", "1", "four", "5",
"2", "2", "1", "1", "2", "-2", "1", "3"
],
"UnitPrice": [
"55000", "750", "1200", "12500", "55000",
"1800", "250", "1200", "1200", "2,500",
"55000", "unknown", "18000", "500000", "750"
],
"Customer": [
"Asha", "Rahul", "Priya", "Kiran", "Meena",
"Arun", "Divya", "Priya", "Priya", "Sana",
"Vikram", "Nisha", "Ravi", "Isha", "Rahul"
]
}
df = pd.DataFrame(data)
df
[3]:
OrderDate Product Quantity UnitPrice Customer
Python 3 (ipykernel) | Idle
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Untitled.ipynb


localhost:8888/lab/tree/Untitled.ipynb
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Untitled.ipynb
Code
Python 3 (ipykernel)
Name
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"55000", "750", "1200", "12500", "55000",
"1800", "250", "1200", "1200", "2,500",
"55000", "unknown", "18000", "500000", "750"
],
"Customer": [
"Asha", "Rahul", "Priya", "Kiran", "Meena",
"Arun", "Divya", "Priya", "Priya", "Sana",
"Vikram", "Nisha", "Ravi", "Isha", "Rahul"
]
}
df = pd.DataFrame(data)
df
[3]:
OrderDate Product Quantity UnitPrice Customer
0 2026-10-01 Laptop 2 55000 Asha
1 2026/10/02 mouse 3 750 Rahul
2 03-10-2026 Keyboard 2 1200 Priya
3 2026-10-04 Monitor 1 12500 Kiran
4 not available Laptop 1 55000 Meena
5 2026-10-06 Headphones four 1800 Arun
6 2026-10-07 USB Cable 5 250 Divya
7 2026-10-08 Keyboard 2 1200 Priya
8 2026-10-08 Keyboard 2 1200 Priya
9 2026-10-10 Webcam 1 2,500 Sana
10 2026-10-11 Laptop 1 55000 Vikram
11 2026-10-12 Desk Chair 2 unknown Nisha
12 2026-10-13 Tablet -2 18000 Ravi
13 2026-10-14 Projector 1 500000 Isha
14 2026-10-15 mouse 3 750 Rahul
Python 3 (ipykernel) | Idle
Mode: Command
Cell 2/2
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Inspect the dataset



```
[4]: df.head()
```

	OrderDate	Product	Quantity	UnitPrice	Customer
0	2026-10-01	Laptop	2	55000	Asha
1	2026/10/02	mouse	3	750	Rahul
2	03-10-2026	Keyboard	2	1200	Priya
3	2026-10-04	Monitor	1	12500	Kiran
4	not available	Laptop	1	55000	Meena

```
[5]: df.tail()
```

	OrderDate	Product	Quantity	UnitPrice	Customer
10	2026-10-11	Laptop	1	55000	Vikram
11	2026-10-12	Desk Chair	2	unknown	Nisha
12	2026-10-13	Tablet	-2	18000	Ravi
13	2026-10-14	Projector	1	500000	Isha
14	2026-10-15	mouse	3	750	Rahul

```
[6]: df.shape
```

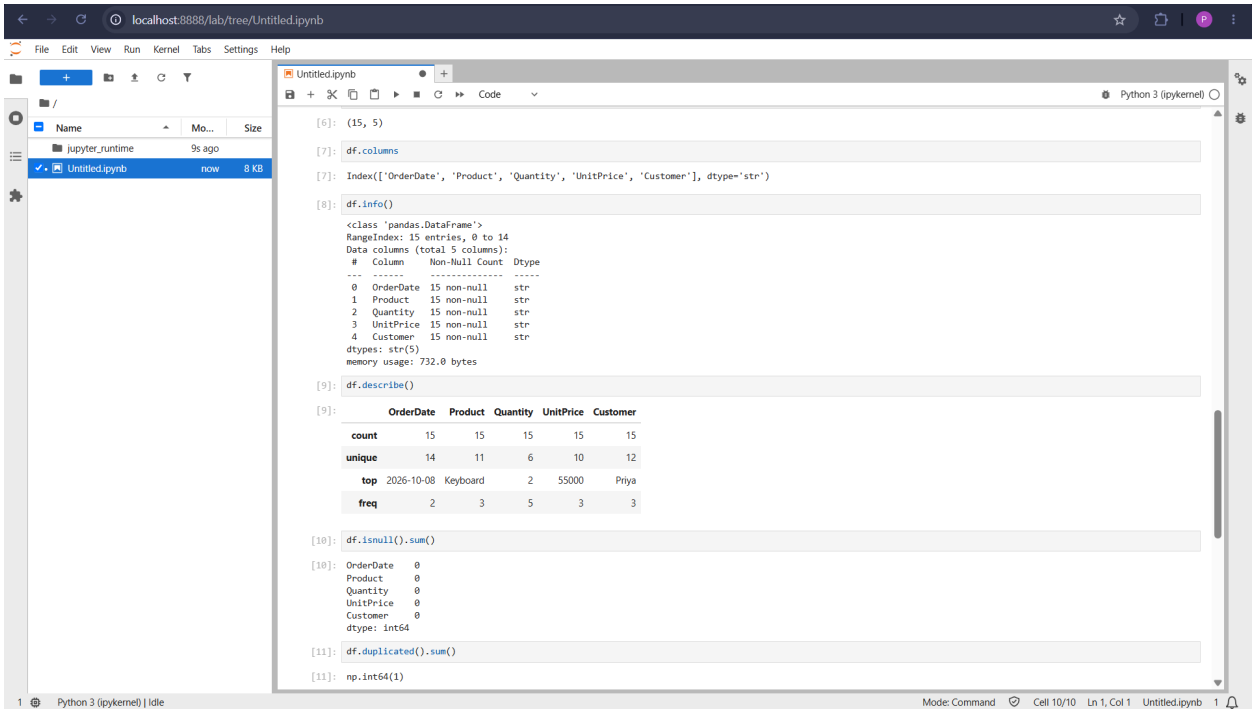
```
[6]: (15, 5)
```

```
[7]: df.columns
```

```
[7]: Index(['OrderDate', 'Product', 'Quantity', 'UnitPrice', 'Customer'], dtype='str')
```

```
[8]: df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 15 entries, 0 to 14
Data columns (total 5 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   OrderDate  15 non-null     str
 1   Product    15 non-null     str
```



```
[6]: (15, 5)
```

```
[7]: df.columns
```

```
[7]: Index(['OrderDate', 'Product', 'Quantity', 'UnitPrice', 'Customer'], dtype='str')
```

```
[8]: df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 15 entries, 0 to 14
Data columns (total 5 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   OrderDate  15 non-null     str
 1   Product    15 non-null     str
 2   Quantity   15 non-null     str
 3   UnitPrice  15 non-null     str
 4   Customer   15 non-null     str
dtypes: str(5)
memory usage: 732.0 bytes
```

```
[9]: df.describe()
```

	OrderDate	Product	Quantity	UnitPrice	Customer
count	15	15	15	15	15
unique	14	11	6	10	12
top	2026-10-08	Keyboard	2	55000	Priya
freq	2	3	5	3	3

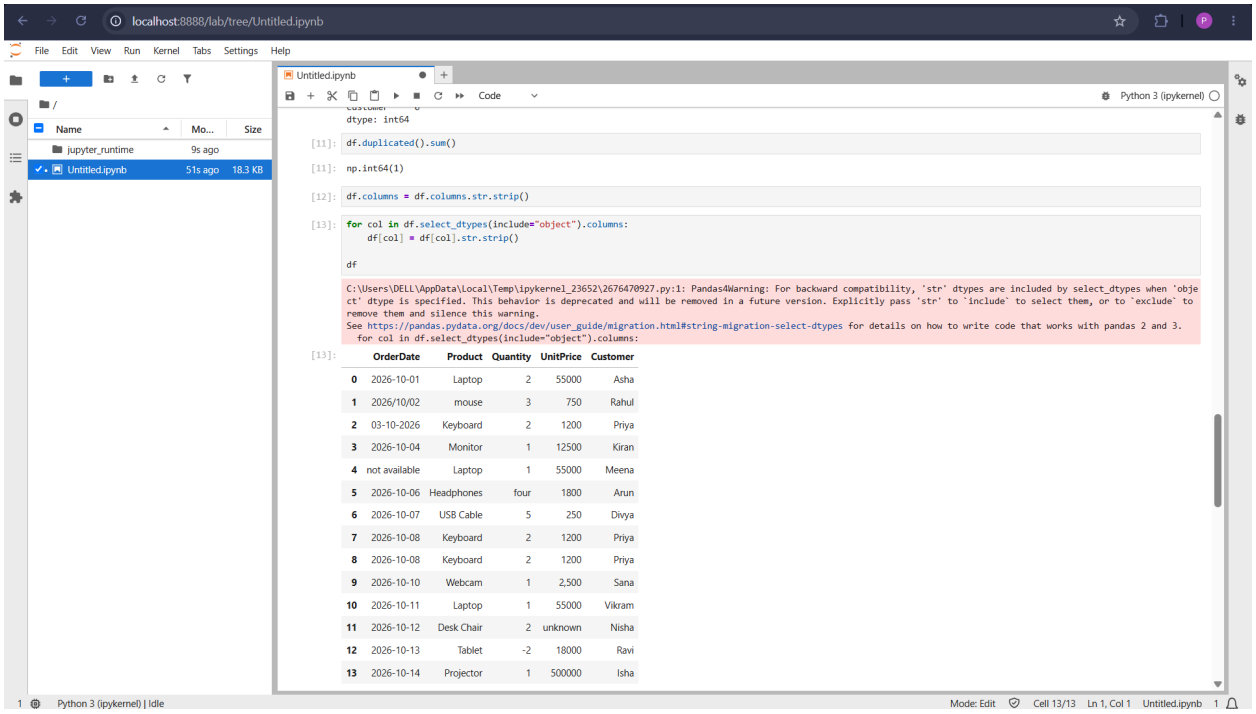
```
[10]: df.isnull().sum()
```

```
[10]: OrderDate    0
      Product      0
      Quantity     0
      UnitPrice    0
      Customer     0
      dtype: int64
```

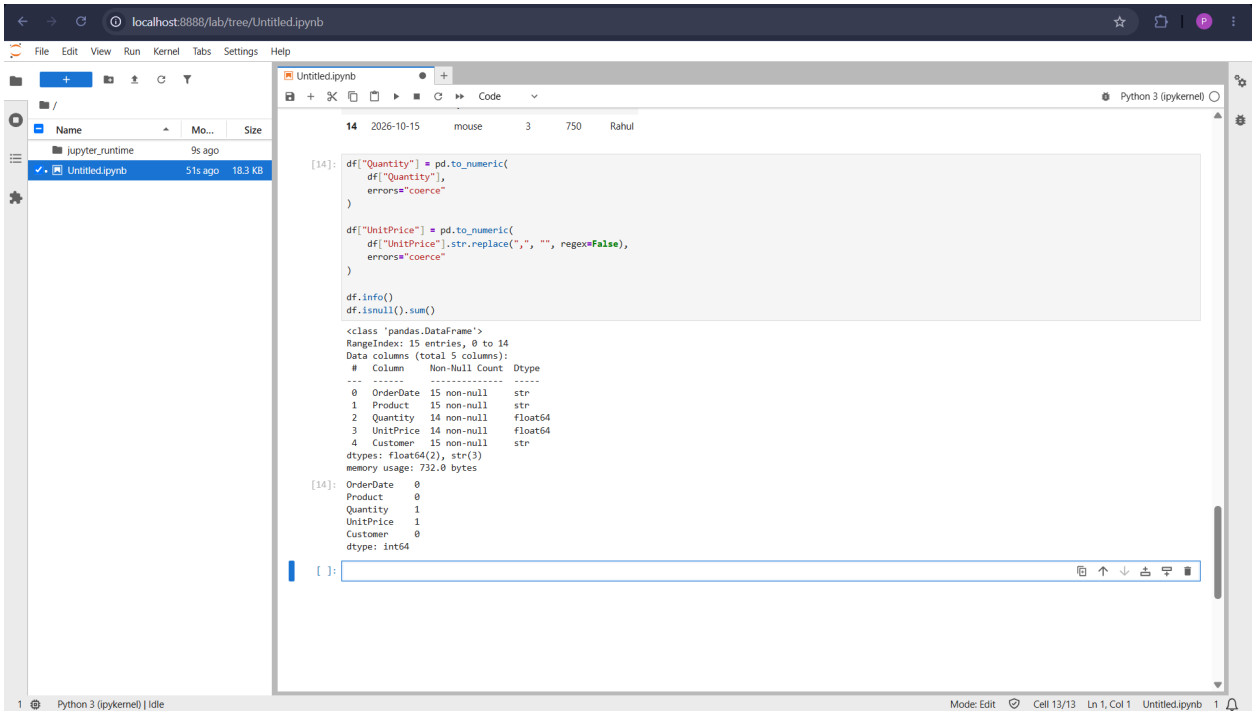
```
[11]: df.duplicated().sum()
```

```
[11]: np.int64(1)
```

Clean the text columns



Convert numeric columns



Convert the date column

[15]:
```
df["OrderDate"] = pd.to_datetime(
    df["OrderDate"],
    errors="coerce",
    format="mixed",
    dayfirst=True
)

df["OrderDate"]
df["OrderDate"].dtype
```

[15]:
```
dtype('<M8[us]')
```

Calculate revenue

[16]:
```
df["Revenue"] = df["Quantity"] * df["UnitPrice"]

df
```

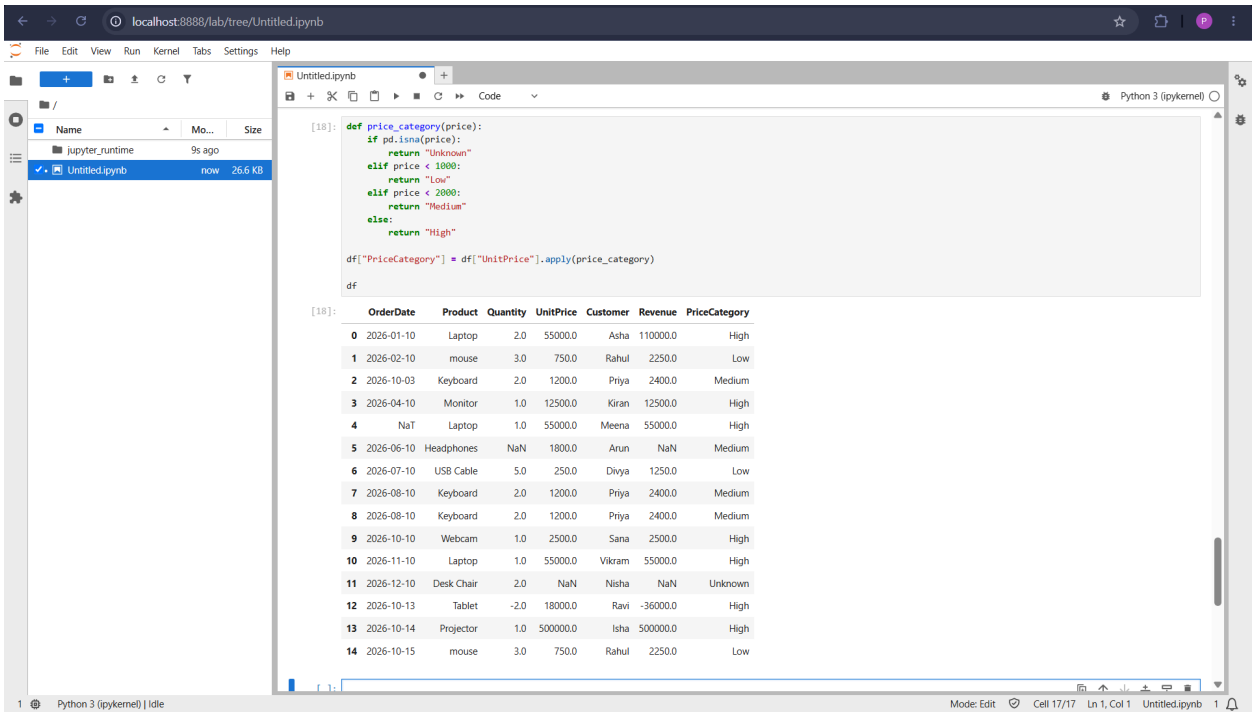
[16]:

	OrderDate	Product	Quantity	UnitPrice	Customer	Revenue
0	2026-01-10	Laptop	2.0	55000.0	Asha	110000.0
1	2026-02-10	mouse	3.0	750.0	Rahul	2250.0
2	2026-10-03	Keyboard	2.0	1200.0	Priya	2400.0
3	2026-04-10	Monitor	1.0	12500.0	Kiran	12500.0
4	NaT	Laptop	1.0	55000.0	Meena	55000.0
5	2026-06-10	Headphones	NaN	1800.0	Arun	NaN
6	2026-07-10	USB Cable	5.0	250.0	Divya	1250.0
7	2026-08-10	Keyboard	2.0	1200.0	Priya	2400.0
8	2026-08-10	Keyboard	2.0	1200.0	Priya	2400.0
9	2026-10-10	Webcam	1.0	2500.0	Sana	2500.0
10	2026-11-10	Laptop	1.0	55000.0	Vikram	55000.0
11	2026-12-10	Desk Chair	2.0	NaN	Nisha	NaN
12	2026-10-13	Tablet	-2.0	18000.0	Ravi	-36000.0
13	2026-10-14	Projector	1.0	500000.0	Isha	500000.0
14	2026-10-15	mouse	3.0	750.0	Rahul	2250.0

[17]:
```
print("Total Revenue:", df["Revenue"].sum())
print("Minimum Revenue:", df["Revenue"].min())
print("Maximum Revenue:", df["Revenue"].max())
print("Average Revenue:", df["Revenue"].mean())
```

```
Total Revenue: 711950.0
Minimum Revenue: -36000.0
Maximum Revenue: 500000.0
Average Revenue: 54765.38461538462
```

Create price categories



Detect outliers using IQR

[19]:
```
Q1 = df["UnitPrice"].quantile(0.25)
Q3 = df["UnitPrice"].quantile(0.75)

IQR = Q3 - Q1

lower_limit = Q1 - 1.5 * IQR
upper_limit = Q3 + 1.5 * IQR

print("Q1:", Q1)
print("Q3:", Q3)
print("IQR:", IQR)
print("Lower Limit:", lower_limit)
print("Upper Limit:", upper_limit)
```

```
Q1: 1200.0
Q3: 45750.0
IQR: 44550.0
Lower Limit: -65625.0
Upper Limit: 112575.0
```

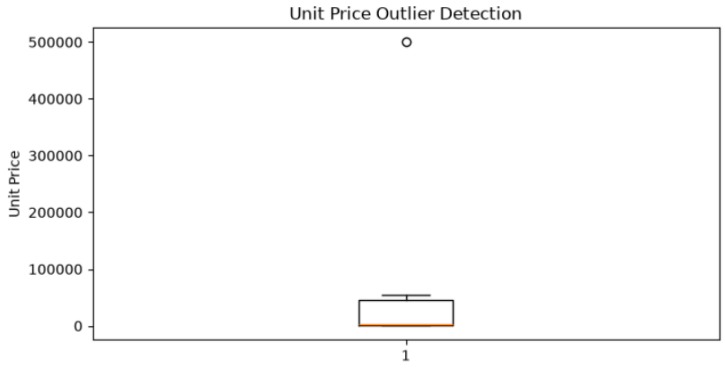
[20]:
```
outliers = df[
    (df["UnitPrice"] < lower_limit) |
    (df["UnitPrice"] > upper_limit)
]

outliers[["Product", "UnitPrice"]]
```

[20]:

	Product	UnitPrice
13	Projector	500000.0

[21]:
```
plt.figure(figsize=(8, 4))
plt.boxplot(df["UnitPrice"].dropna())
plt.title("Unit Price Outlier Detection")
plt.ylabel("Unit Price")
plt.show()
```



Find and remove duplicates

[22]:
```
df[df.duplicated()]
```

[22]:

	OrderDate	Product	Quantity	UnitPrice	Customer	Revenue	PriceCategory
8	2026-08-10	Keyboard	2.0	1200.0	Priya	2400.0	Medium

[23]:
```
df = df.drop_duplicates()

df.shape
df.duplicated().sum()
```

[23]:
```
np.int64(0)
```

Display selected columns

```
[24]: df[["Product", "UnitPrice", "PriceCategory"]]
```

[24]:

	Product	UnitPrice	PriceCategory
0	Laptop	55000.0	High
1	mouse	750.0	Low
2	Keyboard	1200.0	Medium
3	Monitor	12500.0	High
4	Laptop	55000.0	High
5	Headphones	1800.0	Medium
6	USB Cable	250.0	Low
7	Keyboard	1200.0	Medium
9	Webcam	2500.0	High
10	Laptop	55000.0	High
11	Desk Chair	NaN	Unknown
12	Tablet	18000.0	High
13	Projector	500000.0	High
14	mouse	750.0	Low

Final validation

[25]:
```
df.head()
df.shape
df.info()
df.describe()
df.isnull().sum()
df.duplicated().sum()
```

```
<class 'pandas.DataFrame'>
Index: 14 entries, 0 to 14
Data columns (total 7 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   OrderDate      13 non-null     datetime64[us]
 1   Product        14 non-null     str
 2   Quantity       13 non-null     float64
 3   UnitPrice      13 non-null     float64
 4   Customer       14 non-null     str
 5   Revenue        12 non-null     float64
 6   PriceCategory  14 non-null     str
dtypes: datetime64[us](1), float64(3), str(3)
memory usage: 896.0 bytes
```

[25]:
```
np.int64(0)
```